Do pre-war buildings still sell at a premium? Evidence from 845,607 New York City property sales

Group name	Team Amigos
Dataset	data.cityofnewyork.us, dataset w2pb-icbu (NYC Department of Finance, *NYC Citywide Annualized Calendar Sales Update*)
Code	github.com/ami4go/Data_Science_Project_NYC_Property_Sales (scripts, README, frozen data link)

#	Name	Roll no.	Institute email	Contribution
1	Raghav	2024450	raghav24450@iiitd.ac.in	Dataset, raw audit and size check; README (1)
2	Amit	2024064	amit24064@iiitd.ac.in	Cleaning, multi-lot rule and the 70/30 split; before and after (2, 3)
3	Harsh	2024236	harsh24236@iiitd.ac.in	EDA figures and problem statements (4)
4	Bhudeep	2024154	bhudeep24154@iiitd.ac.in	Test plan; Tests 1–2: pre-war premium and mansion tax (5)
5	Parth	2024406	parth24406@iiitd.ac.in	Tests 3–4: transfers and mixed use; limitations and report (5, 6)

1. Dataset

Source: NYC Department of Finance, *NYC Citywide Annualized Calendar Sales Update*, NYC Open Data (dataset `w2pb-icbu`), downloaded on 20 October 2026 and frozen, because the city keeps updating it (SHA-256 `cf8d76c1…`; the code refuses any other file). One row is one recorded sale: a deed on one tax lot, dated 1 Jan 2016 to 31 Dec 2025.

What is raw about it: 31.0% of the "sales" are $0 ownership transfers; floor and land area are stored as text with commas (`"55,000"`) and are 0 or blank in 44.3% of rows; one building category is spelled two ways (91 labels for 47 types); the borough is a code from 1 to 5; the `EASE-MENT` column is entirely blank; 3,386 rows are exact duplicates; and a deal covering several lots repeats its *total* price on every lot. Prices run from $0 to $2.4 billion.

	Raw file	After encoding
Instances (rows)	845,607	566,956 market sales
Attributes (columns)	29	71
Instances × attributes	24,522,603	40,253,876

Size check: $566{,}956 \times 71 = 40{,}253{,}876 \geq 3{,}000{,}000$ (13.4 times the minimum). The raw table already clears 3M; one-hot encoding borough (5 levels), building category (47) and tax class (4) adds 53 columns to 15 numeric and derived ones. Even our smallest test population, the 248,409 sales behind P1, gives $248{,}409 \times 71 = 17.6$M.

Key features

Feature	Type	Real-world meaning
`sale_price`	continuous	recorded price, US$; response for P2–P4
`price_per_sqft`	continuous	derived: price / gross floor area; response for P1
`gross_sqft`	continuous	floor area of the building, sq ft
`year_built`	discrete	assessor's year built; pre-war = before 1940
`total_units`	discrete	residential + commercial units on the lot
`borough`	nominal	5 boroughs
`neighborhood`	nominal	262 assessor neighbourhoods
`building_category`	nominal	47 types, e.g. one-family, condo elevator apartment
`latitude`, `longitude`	continuous	location; used for the toll-zone flag
`sale_date`	date	2016-01-01 to 2025-12-31

2. Data preparation

Issue found	What we did	Why, and not the alternative
$0–$100 sales (274,201; 32.4%)	flagged *nominal*; kept only for the transfer test (T3)	transfers between relatives or companies are not prices; deleting them would make T3 impossible
Area stored as text (`"55,000"`)	commas removed, converted to numbers	a format fix; nothing is lost
Floor area 0 or blank (374,853; 44.3%)	0 set to missing; **not imputed**	missing by design: 95% of condo/co-op unit sales lack it vs 5% of others (Fig. 3); imputing would invent P1's denominator

Issue found	What we did	Why, and not the alternative
91 category labels for 47 types	spaces collapsed, upper-cased	otherwise one-hot encoding splits one type into two columns
3,386 exact duplicate rows	dropped	a sale counted twice inflates counts and shrinks p-values
Multi-lot deals (35,044 rows): one deed repeats its total price on every lot	flagged; no price per sq ft; price statistics use one row per deal (566,956 rows = 545,669 deals)	a null model separates real deals from same-day coincidences (Fig. 1); counting lots instead of deals flips the toll-zone comparison (Fig. 6)
Year built 0 or before 1800 (63,355)	set to missing; excluded from P1 only	placeholders and typos; the sales are valid for other questions
Year built heaped on round years (Fig. 4)	kept; T1 is repeated without 1935–1945	dropping rounded years would remove most pre-war stock
Price per sq ft outliers (3,833)	flagged outside $3\times$IQR fences on log $/sq ft per building type, **fitted on the exploration set only**	per type because levels differ; fitted like a scaler, so the confirmation set is not looked at
Prices above $100M (1,083)	kept; analysed on a log scale	real trophy and portfolio deals; trimming them deletes real sales
Toll zone (Manhattan, 60th St and below) is not a column	flag from coordinates, on a line fitted to the rotated street grid	all 24,072 sales on East/West 50th–70th St land on the correct side
Nominal columns	one-hot: borough, building category, tax class	label codes would impose a fake order

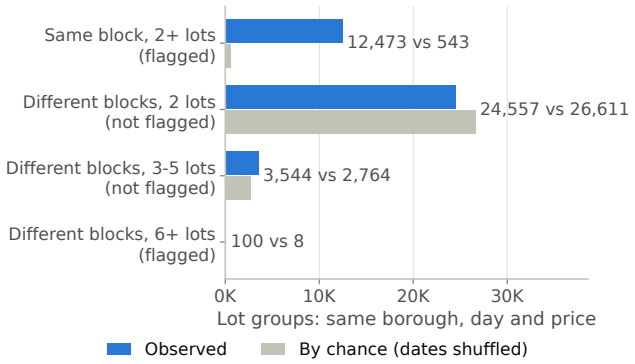


Figure 1: Same-block lot groups sharing a day and price are real multi-lot deals (12,473 observed vs 543 when sale dates are shuffled); two-lot matches across blocks are coincidence (24,557 vs 26,611), so they are not flagged.

3. What changed after preparation

Measure	Before	After
Rows	845,607	842,221
Attributes	29	55 (28 kept + 27 derived or flags)
Exact duplicate rows	3,386	0
Building-category labels	91	47
Numbers stored as text	2 columns	0
Hidden missing (0 meaning "not recorded")	314,567	0 (now explicit)
Market-price rows	571,406 (> $100)	566,956 (> $10,000) = 545,669 deals
Rows with a valid price per sq ft	–	248,627

Missing cells *rise* from 1,793,579 to 2,097,690 (excluding the dropped blank column), because 314,567 zeros that meant "not recorded" are now marked missing; left as 0 they would bias every mean. We lost only the 3,386 duplicate rows outright, but each question can use fewer: price analyses drop the 32.4% transfers, and price per sq ft is defined for 29.4% of raw rows (Fig. 2), so P1 speaks for houses and rental buildings, not apartments.

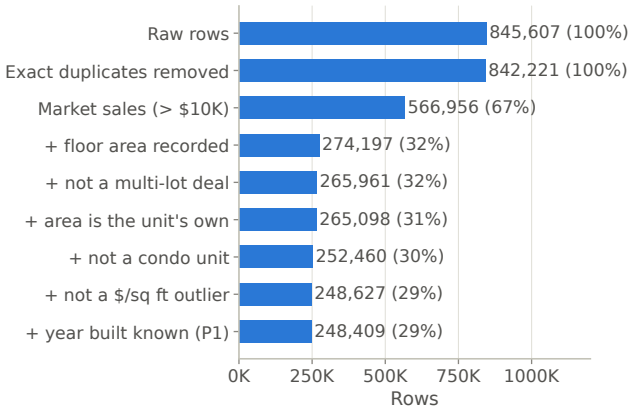


Figure 2: Only 29% of raw rows can answer a price-per-sq-ft question: transfers remove a third, and missing floor area another 35%.

4. Basic questions and problem statement

Split before exploring: 70% exploration / 30% confirmation, **seed 42**, stratified by borough $\times$ sale year (50 strata; exploration share 0.698–0.710 in each), with all lots of one deal in the same half: 590,043 and 252,178 rows. All figures in this section use the exploration set; the tests in Section 5 use only the confirmation set.

Basic questions

1. Is Manhattan the most expensive borough? By median deal price yes ($1.2M vs $514K–$885K), though the price ranges overlap heavily.
2. Do bigger homes cost more? Yes, moderately: Spearman $\rho = 0.49$ for 1–3 family homes.
3. Have prices risen since 2016? Yes, in every borough: +22% in Manhattan, +48–71% elsewhere.
4. Which neighbourhood is priciest per sq ft? Greenwich Village-West, $2,020.

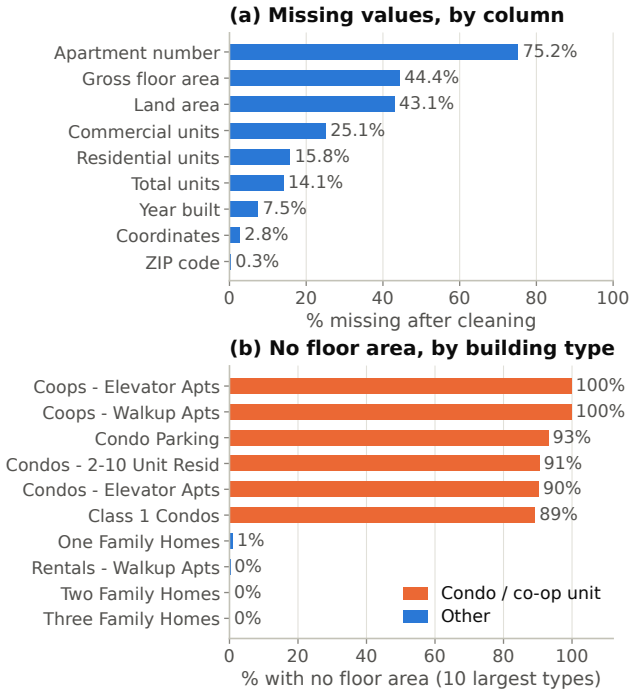


Figure 3: Floor area is missing for 44% of sales, but for 95% of condo/co-op unit sales and only 5% of the rest: missing by design, not at random, so we do not impute it.

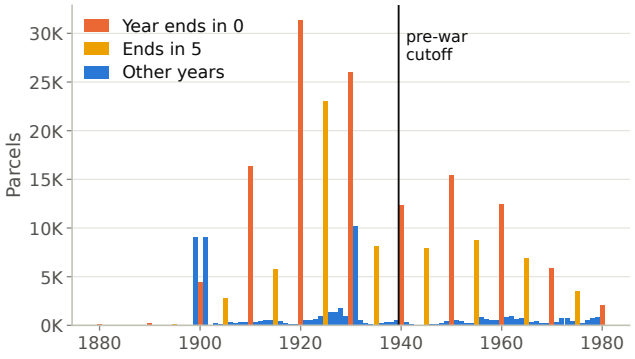


Figure 4: 61% of parcels have a year built ending in 0 or 5 (20% expected); 1940 holds 12,337 parcels vs 540 in 1939, so the pre-war cutoff needs a sensitivity check.

Our problem statements

P1 (primary). Within the same neighbourhood and building type, do pre-war buildings (built before 1940) sell for more per square foot than newer ones?

P2. After the July 2019 mansion-tax step at $2M, do more residential deals land just below $2M than before?

P3. Holding building type fixed, does the share of $0–$100 ownership transfers differ between boroughs?

P4. Within the same neighbourhood and building type, do buildings with shops sell at a different price per square foot from purely residential ones?

How they are novel: the basic questions read one column against price, and with 250K rows any such gap is "significant". Ours compare like with like: the same neighbourhood and type (P1, P4), the same building mix (P3), or the same narrow price window before and after a tax change, with placebo windows where the tax did not change (P2).

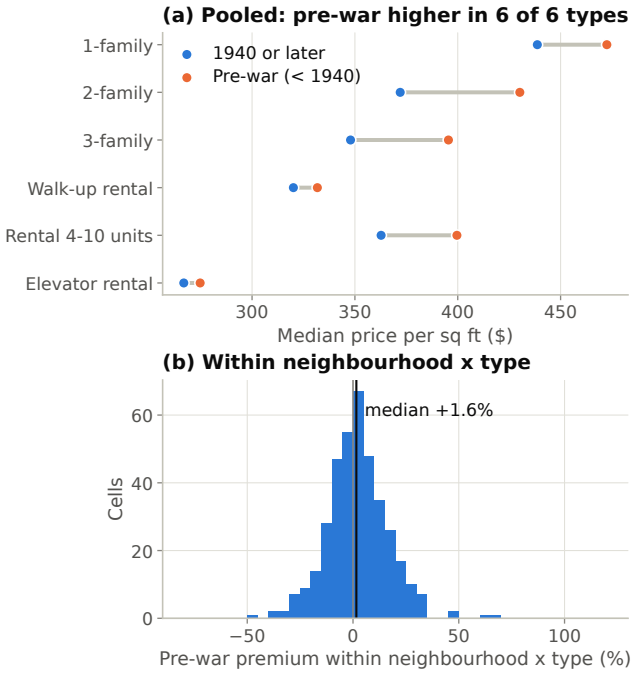


Figure 5: Pooled, pre-war residential buildings sell for 6.6% more per sq ft; within the same neighbourhood and type the median premium is only +1.6% (383 cells, IQR -6% to $+11\%$), near the 5% that would matter.

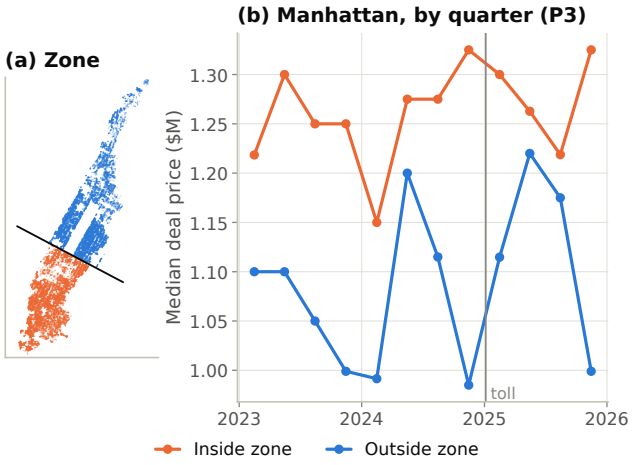


Figure 6: Counting each deal once, 2024–25 median prices rose 1.6% inside the toll zone vs 6.6% outside; counting every lot of multi-lot deals would show $+10.4\%$ vs $+5.1\%$, driven by one $170M deal repeated on 280 lots.

Each needs columns no raw row records (price per sq ft, a pre-war flag, deal-level prices, a price class, a mixed-use flag).

Correlation is not causation: P2 has a causal flavour: the tax pushed prices under $2M. Two things support that reading: the comparison is with the same window before the tax, and windows at $1.5M and $2.5M, where the tax did not step, act as placebos. It still cannot tell a real price cut from money moved off the deed (furniture, closing credits). P1, P3 and P4 are associations: pre-war buildings may also differ in renovation, and mixed-use buildings sit on busier streets.

Hypotheses we dropped: COVID and Manhattan prices (too

few Manhattan per-sq-ft sales), the congestion toll (no stable pre-toll trend; Section 6), new buildings (too few sales), and pooled questions such as tax class by borough or older vs newer citywide, which any test would reject.

5. Statistical inferences

Rows tested: only the confirmation set (252,178 rows), with every hypothesis, test and threshold fixed in `reports/test_plan.md` before the tests were run. All tests are from the course (z, t, paired t, F and χ^2); $\alpha = 0.05$. Fig. 7 shows every estimate against the size that would matter.

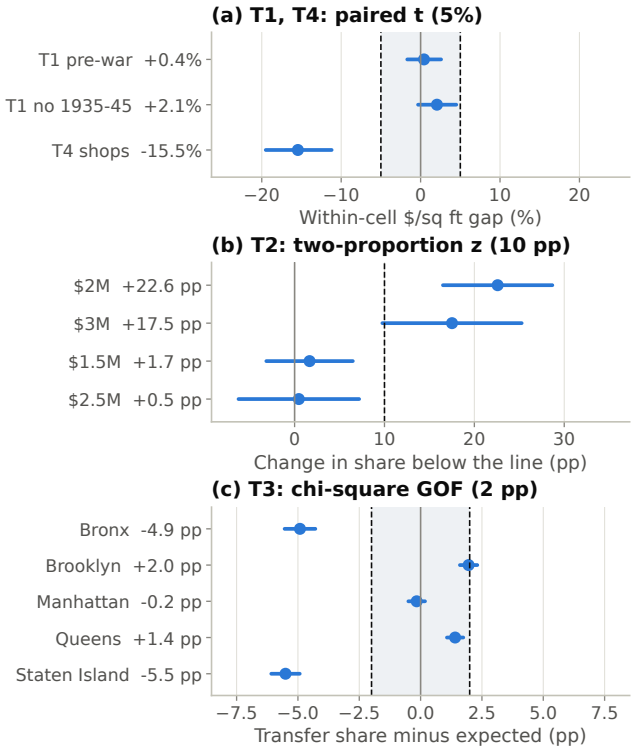


Figure 7: Estimates with 95% CIs; shaded = smaller than the size that matters. Only T2 at $2M, T4, and Bronx / Staten Island in T3 clear it; the pre-war premium (T1) does not.

Test 1 (P1): do pre-war buildings sell for more per sq ft, like for like?

a) Hypotheses, written before running the test

H_0	$\mu_d = 0$
H_1	$\mu_d > 0$
Significance level	$\alpha = 0.05$
Side	one-sided (right)
Test	paired t-test
Smallest difference that matters	5% ($\mu_d = \ln 1.05 = 0.049$)

d = mean log $/sq ft of pre-war minus newer sales in one neighbourhood $\times$ building-type cell (1–3 family and rental buildings; cells with ≥ 5 of each). One-sided, because P1 claims a premium. 5%, because a smaller gap is lost in normal price negotiation.

b) Other tests we considered

Each cell is a matched pair (same place, same type), σ is unknown, and 367 cells make the mean difference near normal: a paired t-test. Pairing removes location, which drives most of the raw gap (Fig. 5).

Also considered	Why we did not use it
Welch two-sample t on all sales	ignores neighbourhood, so it measures location mix. Run anyway: +7.1% (6.3% to 8.0%), $t = 16.75$, d.f. $= 71{,}025$, $p < 10^{-60}$.
Pooled two-sample t	needs equal variances; the F-test rejects them ($F = 1.60$, d.f. $(39{,}876; 31{,}158)$, $p < 0.001$).
z-test for two means	needs the population σ, which we do not know
t-test on the correlation of year built with $/sq ft	sees only a straight-line trend over all years, not the pre-war contrast, and ignores location

c) Result

Estimate	+0.43% (367 cells, 58,139 sales)
95% confidence interval	-1.68% to $+2.58\%$
Test statistic (d.f.)	$t = 0.394$ (366)
p-value	0.35
Decision	do not reject H_0

Like for like, there is no pre-war premium: the whole interval lies inside $\pm 5\%$. The 7.1% pooled gap is location and building mix, not age. Without year built 1935–1945 (the heaping at 1940) the estimate is +2.1% (95% CI -0.3% to $+4.5\%$, $t = 1.70$, d.f. 346, $p = 0.045$): borderline significant, still below what matters.

Test 2 (P2): did the 2019 mansion tax push deals just below $2M?

a) Hypotheses, written before running the test

H_0	$p_{\text{post}} = p_{\text{pre}}$
H_1	$p_{\text{post}} > p_{\text{pre}}$
Significance level	$\alpha = 0.05$
Side	one-sided (right)
Test	z-test for two proportions (pooled)
Smallest difference that matters	10 percentage points

p = share of residential deals (tax class 1–2, one row per deal) within 2.5% of $2M that are below $2M; pre = 2016–18, post = 2020–25, 2019 dropped (contracts signed before April kept the old rules). Since July 2019 crossing $2M raises the tax on the whole price (about $5,000), and the tax only rewards being below, so the test is one-sided. 10 pp is about one buyer in ten moving their price for the tax.

b) Other tests we considered

Two independent groups of deals, a yes/no outcome and counts in the hundreds (all four cells ≥ 15): a z-test for two proportions, with the rate pooled because H_0 says both periods share one.

Also considered	Why we did not use it
χ^2 test of independence (2×2)	same evidence but two-sided only. Run anyway: $\chi^2 = 54.85 = z^2$, d.f. 1, $p < 10^{-12}$.
z-test for one proportion vs 50%	50/50 is arbitrary: round prices at exactly $2,000,000 make the pre-tax split uneven, so the pre-period is the right baseline

c) Result

Estimate	47.0% (157/334) → 69.6% (673/967): +22.6 pp
95% confidence interval	+16.5 to +28.7 pp
Test statistic	$z = 7.41$
p-value	6.5×10^{-14}
Decision	reject H_0

After the tax, deals pile up just below $2M, and the whole interval is above the 10 pp that matters. The placebos behave as they should: at $1.5M (+1.7 pp, $p = 0.25$) and $2.5M (+0.5 pp, $p = 0.45$), where the tax did not change, nothing moved. At $3M, the next tax step, the share rose 17.5 pp (95% CI +9.8 to +25.3, $z = 4.43$, $p < 10^{-5}$).

Test 3 (P3): do boroughs differ in $0 transfers once building type is fixed?

a) Hypotheses, written before running the test

$\boldsymbol{H_0}$	$O_b = E_b$ for every borough
$\boldsymbol{H_1}$	at least one borough differs
Significance level	$\alpha = 0.05$
Side	upper tail of χ^2
Test	χ^2 goodness of fit, d.f. $= 5 - 1 = 4$
Smallest difference that matters	2 pp from a borough's expected share

O_b = $0–$100 transfers in borough b; $E_b = \sum_c r_c\, n_{bc}$, where r_c is the citywide transfer rate of building category c and n_{bc} the deeds of that category in borough b. So E_b is what the borough's building mix alone predicts.

b) Other tests we considered

The data are counts in five categories compared with a predicted split: goodness of fit. Building E_b from category rates is what holds building type fixed.

Also considered	Why we did not use it
χ^2 test of independence, borough × transfer	mixes building type into the borough effect. Run anyway: $\chi^2 = 4{,}473$, d.f. 4, raw shares 21.1% (Manhattan) to 37.7% (Brooklyn), a 16.6 pp spread.
z-tests for two proportions, pair by pair	10 pairs means 10 more tests and 10 more chances of a false alarm
χ^2 test of independence inside each building category	holds type fixed too, but means 47 separate tests and many categories with tiny counts

c) Result

Test statistic (d.f.)	$\chi^2 = 461.3$ (4); smallest $E_b = 7{,}873$
p-value	$< 10^{-90}$
Decision	reject H_0
Observed minus expected	Bronx -4.9 pp (-5.5 to -4.3), Staten Island -5.5 (-6.1 to -4.9), Brooklyn $+2.0$ ($+1.6$ to $+2.3$), Queens $+1.4$ ($+1.1$ to $+1.7$), Manhattan -0.2 (-0.5 to $+0.2$)

Holding building type fixed changes the story. Manhattan's low raw share (21.1%) is fully explained by what sells there (condos and co-ops): it sits 0.2 pp from its expectation. The gaps that matter are in the Bronx and Staten Island, with about 5 pp *fewer* transfers than their building mix predicts. Expected counts use rates from the same data, so d.f. $= 4$ is approximate.

Test 4 (P4): do buildings with shops sell at a different price per sq ft?

a) Hypotheses, written before running the test

$\boldsymbol{H_0}$	$\mu_d = 0$
$\boldsymbol{H_1}$	$\mu_d \neq 0$
Significance level	$\alpha = 0.05$
Side	two-sided
Test	paired t-test
Smallest difference that matters	5% either way

d = mean log $/sq ft of buildings with commercial units minus purely residential ones, per neighbourhood × type cell (≥ 3 of each; mixed-use sales are rare). Two-sided: shops can add rent (premium) or noise and traffic (discount).

b) Other tests we considered

The same matched design as Test 1: one pair per cell, σ unknown, 225 cells.

Also considered	Why we did not use it
Welch two-sample t on all sales	mixed-use buildings sit on commercial streets in other neighbourhoods. Run anyway: -9.3% (-11.9% to -6.7%), $t = -6.79$, d.f. 3,370.
Pooled two-sample t	the F-test rejects equal variances ($F = 2.23$, $p < 0.001$)
t-test on the correlation of commercial units with $/sq ft	linear only, and most buildings have 0 or 1 commercial unit

c) Result

Estimate	-15.5% (225 cells, 41,093 sales)
95% confidence interval	-19.5% to -11.2%
Test statistic (d.f.)	$t = -6.74$ (224)
p-value	1.3×10^{-10}
Decision	reject H_0

Buildings with shops sell for about 15% less per sq ft than purely residential buildings of the same type in the same neighbourhood, and the whole interval is beyond 5%. Ignoring neighbourhood understates it (-9.3%), because mixed-use buildings are more common in pricier areas.

6. Limitations

- **Association, not cause**, for P1, P3 and P4. In P4, gross floor area includes the shop space, which may simply be valued lower than homes. P2's placebos support a tax effect, but cannot rule out money moved off the deed.
- **The toll question was not tested.** Before the toll, the yearly gap between inside and outside the zone already swung from $+9.2$ to -3.6 pp (2021–24), so 2025's -5.0 pp cannot be told apart from past movement; a before/after test would wrongly credit the toll.
- **Price per sq ft covers houses and rental buildings only** (29.4% of raw rows); condo and co-op units carry no unit area, so P1 and P4 say nothing about apartments.
- **Year built is an assessor estimate**, heaped on round years (Fig. 4); dropping 1935–1945 moves T1 from +0.4% to +2.1%, so "pre-war" carries measurement error.
- **Sales are not independent.** Sales in one building, neighbourhood or month share conditions. Tests 1 and 4 work on cells to soften this; Tests 2 and 3 treat deals as independent, so their p-values are somewhat optimistic.
- **Judgement calls.** The $100 and $10,000 cut-offs, the multi-lot rule, the 2.5% window and the minimum cell sizes were fixed before testing but are still choices.
- **Only properties that sold** are in the data; they may differ from the housing stock as a whole.